

CORE 2B

Motion in 2D - Projectile Motion

Changed-demand transfer: commit first; the protected DECIDE move appears only after the attempt page.

Transfer rule

Every task uses prior exposure from Core1A, Core1B, and Core2A. On the first page of each task, write your model/representation, the decision that justifies it, and your first executable relation. Do not turn the page until you have committed.

Transfer 1

A particle follows $x=2t^2+3t$ and $y=12t-t^2$. At $t=1$ s, decide the direction of motion. Neither velocity component is constant.

Prior exposure: Core1A E1 -> Core1B E1 -> Core2A Practice 1

Safe pre-attempt support

List the givens, event boundaries, and requested quantity. Do not choose the governing extension until you commit.

COMMIT: model / decision / first relation

Changed demand

Both component position laws vary with time. The representation decision must be made before any angle calculation.

Protected DECIDE move

Choose derivative components, not position ratios, to represent instantaneous direction.

Complete reasoning route

1. REPRESENT - List $x(t)$, $y(t)$, and $t=1$ s without selecting an angle formula.
2. DECIDE - Differentiate both position laws.
3. TRANSFORM - $v_x=4t+3=7$; $v_y=12-2t=10$ at $t=1$ s.
4. VERIFY - Velocity is tangent to the path.

Answer

$\theta = \tan^{-1}(10/7)$ above $+x$ (about 55 degrees).

Independent check

A position-vector angle would answer a different question.

Transfer 2

A projectile returns to level ground with speed U . After impact it retains 36% of its kinetic energy, and its rebound velocity has vertical:horizontal component ratio 3:4. Find its next maximum height in terms of U and g .

Prior exposure: Core1A E2 -> Core1B E2 -> Core2A Practice 2

Safe pre-attempt support

List the givens, event boundaries, and requested quantity. Do not choose the governing extension until you commit.

COMMIT: model / decision / first relation

Changed demand

The rebound direction is given as a component ratio rather than an angle.

Protected DECIDE move

Convert the energy fraction to a speed-squared fraction before using the 3:4 component ratio.

Complete reasoning route

1. REPRESENT - Keep retained energy and 3:4 direction information separate.
2. DECIDE - $v_{\text{after}}^2 = 0.36U^2$.
3. TRANSFORM - $v_y = (3/5)v_{\text{after}}$, so $v_y^2 = (9/25)(0.36U^2) = 81U^2/625$.
4. VERIFY - Use $H = v_y^2/(2g)$ and check the strong reduction.

Answer

$$H = 81U^2/(1250g).$$

Independent check

The result includes both the energy fraction and the squared vertical component fraction.

Transfer 3

A projectile reaches its apex at horizontal position L and height H with horizontal speed V . Immediately after the apex, downward gravity changes to $g/4$. Decide the additional horizontal distance before it reaches the launch level.

Prior exposure: Core1A E3 -> Core1B E3 -> Core2A Practice 3

Safe pre-attempt support

List the givens, event boundaries, and requested quantity. Do not choose the governing extension until you commit.

COMMIT: model / decision / first relation

Changed demand

The apex state is supplied directly; only the second-stage displacement is requested.

Protected DECIDE move

Use the supplied apex as the initial state of a new constant-acceleration stage.

Complete reasoning route

1. REPRESENT - Record apex state (L,H,V,0) and post-apex gravity $g/4$.
2. DECIDE - Start stage 2 at the apex.
3. TRANSFORM - $H=0.5(g/4)t^2$ gives $t=2\sqrt{2H/g}$.
4. VERIFY - Weaker gravity must lengthen the descent.

Answer

Additional horizontal distance = $2V\sqrt{2H/g}$.

Independent check

The result is longer than the ordinary- g descent distance from the same apex.

Transfer 4

From the origin, can a finite-speed gravity-only projectile launched exactly along the straight line joining the origin to $P=(5\text{ m}, 1\text{ m})$ pass through P? Justify without solving for a numerical speed.

Prior exposure: Core1A E4 -> Core1B E4 -> Core2A Practice 4

Safe pre-attempt support

List the givens, event boundaries, and requested quantity. Do not choose the governing extension until you commit.

COMMIT: model / decision / first relation

Changed demand

This is a reachability-boundary decision rather than an ordinary solve-for-speed problem.

Protected DECIDE move

Compare P with the straight-line term before attempting to solve for u.

Complete reasoning route

1. REPRESENT - $\tan(\theta)=1/5$, so at $x=5$ the straight launch ray gives $x \tan(\theta)=1$.
2. DECIDE - In the projectile trajectory, gravity subtracts a strictly positive term for finite u.
3. CONNECT - Therefore actual y at $x=5$ is always less than 1 m for finite u.
4. VERIFY - As u tends to infinity the gravity term tends to zero, so the ray is approached only as a limit.

Answer

No finite launch speed can make the projectile pass through P.

Independent check

The infinite-speed limit explains why the point is approached but never reached for finite u.

Transfer 5

A projectile's measured horizontal velocity halves every τ seconds in a medium. Decide whether $x = v_{x0} t$ is valid and write a horizontal displacement expression in terms of v_{x0} , τ , and t consistent with linear drag.

Prior exposure: Core1A E5 -> Core1B E5 -> Core2A Practice 5

Safe pre-attempt support

List the givens, event boundaries, and requested quantity. Do not choose the governing extension until you commit.

COMMIT: model / decision / first relation

Changed demand

The drag parameter is supplied through a velocity half-life instead of c/m .

Protected DECIDE move

Translate the halving statement into an exponential velocity law before computing distance.

Complete reasoning route

1. REPRESENT - $v_x = v_{x0} 2^{-(t/\tau)} = v_{x0} \exp[-(\ln 2)t/\tau]$.
2. DECIDE - $x = v_{x0} t$ is not exact because v_x is not constant.
3. TRANSFORM - Integrate the exponential velocity from 0 to t .
4. VERIFY - Take the short-time limit.

Answer

$$x = (v_{x0} \tau / \ln 2) [1 - 2^{-(t/\tau)}].$$

Independent check

For small t , the expression approaches $v_{x0} t$.